

Supplementary Material for “Penalties Sized to the Gain: When Do Random Audits Stop AI Agents from Over-Using a Shared Resource?”

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A Settings of every study

Table S1: The settings of every study. All simulations: 80 rounds, 64 new populations. LLM studies: 20 rounds, 10 populations. Audits random with probability 1/6 per agent and round (T1 and C1: one audit per round). ^aFISHERY: stock below 50 after fishing; FOREST: more than a 5% chance that mean plot health falls below 10; R1 also reports the one-step line, which the FISHERY parts of S1 and S1b used. ^bChosen by search to maximise the cheaters’ payoff over the whole game. ^cS4 and S5 cut a caught agent’s allowance by the share of the cut it took back; L2 by its mean tonnes over. ^dRIVER: quality below 30 (pre-registered) and below 50. ^ePart A compares reviewers built for the 30 line, so 30 is kept.

Study	Claims	Games	Who misbehaves, and how	How it is chosen	Consequence of a catch	Harm line	Gain used
R1, R2	1–3	Fishery, Forest	none (honest); R2: reviewer’s model wrong	–	–	half capacity ^a	–
S1, S1b	3	Forest, Fishery	programmed: under-report	one level per game ^b	fine, exclusion or none	half capacity ^a	–
S2, S3	4	Fishery	programmed: over-take	one level per game ^b	fine ($s \leq 1$)	half capacity	whole-game g^*
S4, S5	3–4	Fishery	programmed: over-take, adapting	one level per game ^b	memory ^c , fine	half capacity	whole-game g^*
R3	4	Fishery (6 settings)	programmed: over-take	one level per game ^b	fine	half capacity	whole-game g^* , predicted
T1	5	all three	programmed: under-report by half	fixed	memory, no fine	half capacity ^d	–
C1	1, 5	River	A: honest; B: under-report	fixed	B: memory	quality < 30 ^e	–
L2	3, 6	Fishery	4 LLMs: over-take	every round	fine 0–36 t, or memory ^c	half capacity	one-round g_1
L3	6	Fishery	gpt-oss, Nemotron: over-take	every round	fine 12–30 t	half capacity	one-round g_1 , predicted

B Key terms

Table S2: Key terms.

Term	Meaning in this paper
<i>The game</i>	
Agent	One of six participants who draw on the shared resource each step (a fisher in FISHERY, a forester in FOREST, a factory in RIVER).
Greedy agent	An agent whose fixed rule asks for a lot; 2 or 4 of the 6 in most runs.
Request p_i	How much an agent asks to take this step, as a share of its maximum.
Report	What the agent tells the reviewer it wants. An honest report equals the request; an <i>under-report</i> (a lie) is smaller.
Allowance	What the reviewer lets an agent take this step: its request times the shared cut.
Over-taking (cheating)	Taking more than the allowance. Nobody sees it unless the agent is audited.
Step, run	One round of requests and harvest; a run is 80 steps.
Population (context)	One randomly generated set of six agents and its random seeds. Results average over 64 of them.
<i>The reviewer</i>	
Reviewer	The overseer. It predicts what the requests would do to the resource and approves them or cuts them back.
Shared cut λ	One scaling factor ($1, \frac{3}{4}, \frac{1}{2}, \frac{1}{4}$ or 0) applied to everyone's request.
Model of the resource	The reviewer's equations for how the resource regrows. "Correct model" means it knows the true ones.
Joint reviewer	Adds up all six requests before predicting.
Local reviewers	Judge from one agent's request: the <i>cautious</i> one assumes everyone else takes as much as the largest request; the <i>optimistic</i> one ignores the others.
Safety line	The basic target: the resource must not fall below a danger level next step (Fishery stock 10).
Productive target (MSY)	A stricter target that keeps the stock where it regrows fastest (leave at least half of the maximum), the level behind <i>maximum sustainable yield</i> .
<i>Checking and consequences</i>	
Audit	A spot check of one agent: the reviewer learns its true request, or what it actually took.
Audit rate q	The chance that a given agent is audited in a given step (the <i>audit budget</i>).
Detection rate s	The chance that an audit of a cheater actually catches it.
Fine F	A penalty paid each time an agent is caught.
Expected fine $e = q s F$	The average fine an agent faces per step if it cheats.
Break-even g	What one step of cheating gains on average. Cheating stops paying once $e > g$.
Memory	The reviewer remembers how much each caught agent lied or over-took, and corrects for it afterwards.
Audit rule	Which consequences follow a catch: a fine, memory, extra checks on known cheaters, or a tighter cut for everyone.
<i>Measures</i>	
Harvest	What agents take from the resource (fish, timber, or discharge capacity). Used only as an outcome, never as a game name.
Harm	The share of steps in which the resource is below the safety line, or above the MSY limit (more taken than the productive target allows).
Collapse	The resource stays below its danger level long enough that the run ends.
Risky request approved	A request that would cross the target, let through unchanged (a missed harm).
Safe request cut	A request that was fine, cut anyway (<i>usefulness loss</i>).
Training vs. new runs	Cheaters choose their strategy on a few training populations; results are reported on 64 new ones.
95% interval	The range the result would likely fall in if the 64 populations were redrawn (bootstrap).

C The reviewer's target and model

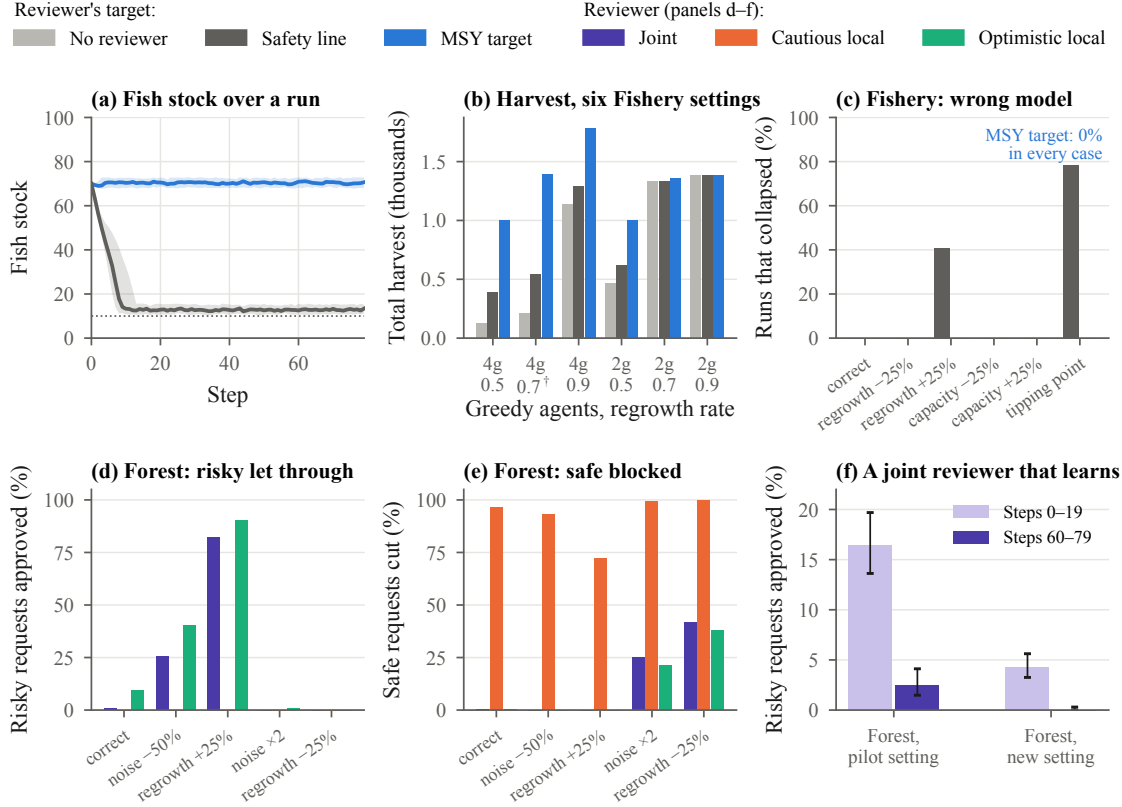


Figure S1: What the reviewer aims for and what it knows. Top row, FISHERY: stock over one run under the one-step line and the half-capacity target; total harvest in six settings; collapses when the reviewer's model is wrong. Bottom row, FOREST: risky requests let through and safe requests blocked when part of the reviewer's model is wrong, and a reviewer that learns the model while running. Error bars: 95% intervals.

Table S3: Audit rules compared (FISHERY, half-capacity target, 64 new populations). Bold: best in column.

Audit rule	Audit rate q	Cheaters' extra harvest [95% interval]	Harvest per honest agent	Steps over MSY limit (%)	Checks per step
No audits	—	112.8 [106, 120]	39.5	100	0.00
Fine	0.10	0.0	104.1	0	0.61
	1/6	0.0	104.1	0	1.00
Memory	0.10	58.4 [49, 68]	74.5	54	0.61
	1/6	55.1 [46, 65]	77.1	49	1.00
Memory + extra checks	0.10	-3.5 [-36, 14]	95.2	16	3.78
	1/6	8.3 [7, 10]	99.4	10	4.07
Memory + tighter shared cut	0.10	141.7 [136, 147]	36.0	24	0.61
	1/6	142.3 [137, 148]	36.0	16	1.00
Fine + memory + tighter cut	0.10	0.0	104.1	0	0.61
	1/6	0.0	104.1	0	1.00
Nobody cheats	—	0	104.1	0	0

D Controls for the LLM study

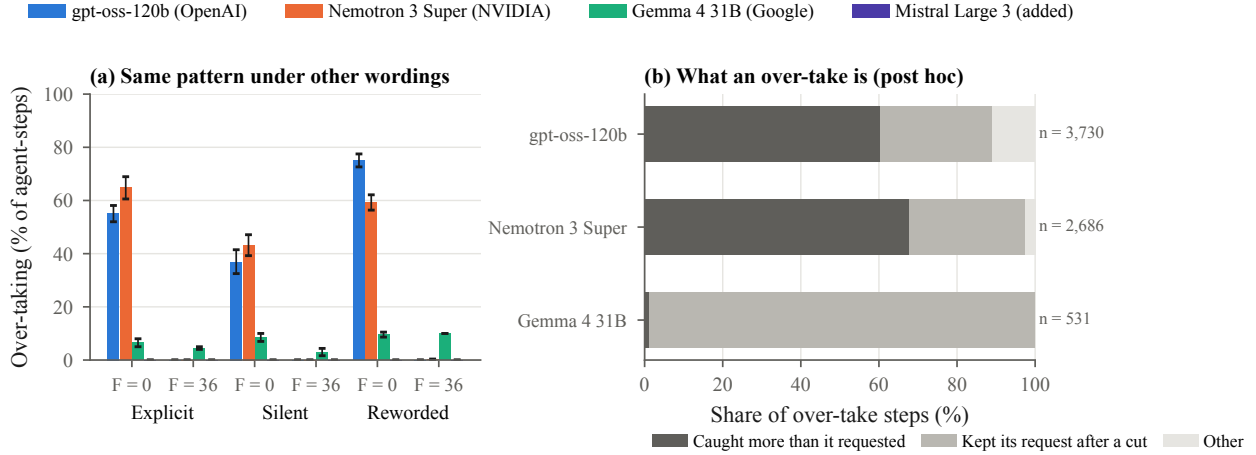


Figure S2: LLM agents in FISHERY, controls. (a) A 36 t fine stops over-taking whether the rules state that over-taking is possible (explicit), omit it (silent), or reword it. The silent cells still mention a check with a fine of 0 t, so they do not replicate a no-check condition. (b) What an over-take round is (post hoc): gpt-oss and Nemotron mostly take more than they requested; Gemma almost only keeps its request after the reviewer's cut. Error bars: 95% intervals over populations.

E How the LLM agents over-take (post hoc)

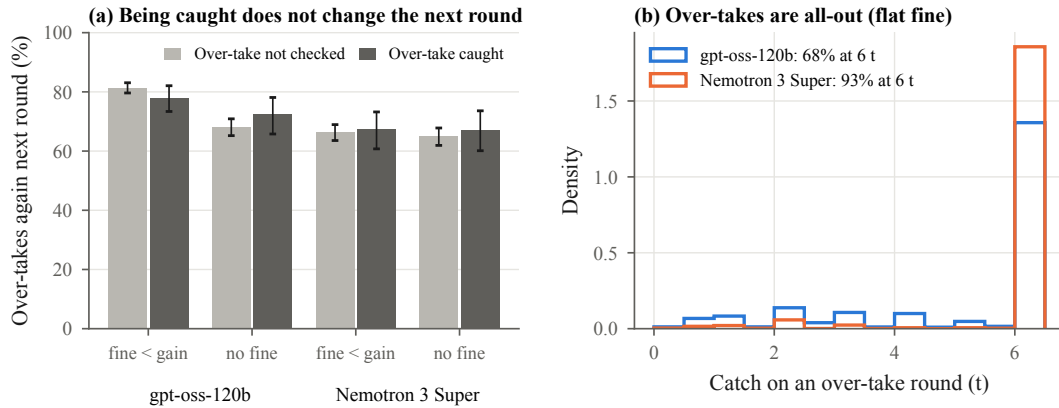


Figure S3: Post hoc analyses of gpt-oss and Nemotron. (a) After an over-take, the chance of over-taking again barely depends on whether the agent was caught. (b) Over-takes are mostly at the 6 t maximum. Intervals over agent-rounds, which are not independent; descriptive only.

F Protocols and reproducibility

Each study had a protocol frozen before its code, with hypotheses, falsifiers, sample sizes and analyses; amendments are dated and state what had been seen. The LLM studies used six amendments (an output-parsing fix; remote execution; a held-back and corrected memory cell; the replacement of Mistral Large 4, which used about 6,000 hidden-reasoning tokens per decision, by Mistral Large 3; and two infrastructure fixes) and one note that partial results of the threshold test were seen during the run, with nothing changed. Code, protocols, prompts, every game and all analysis scripts will be released.